

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science and Engineering

Mid-Semester Examination, July-December 2025

Program & Dept.: B.Tech/B.Tech-M.Tech-DD, CSE | **Semester- III**

Course Name: Object Oriented Programming | **Course Code: CS34105**

Maximum Time: 2 hours | **Max. Marks: 30**

Instructions:

1. *Attempt all questions. Write a concise and precise answer.*
2. *Assume any suitable data, if necessary.*
3. *The marks, Course Outcome (CO), and Bloom's Level (BL) related to questions are mentioned at the end of each question.*

Q1 a) Explain the role of *this* keyword in Java. How is it used to solve ambiguity between instance variables and method parameters? Additionally, describe two other important uses of the *this* keyword with examples. [4 M][CO2][BL1,BL3]

b) Write a Java program to reverse a string without using built-in library function reverse(). [3 M][CO2][BL4]

c) How does JIT (Just-In-Time compiler) within the JVM improve program performance? Where does JIT fit in the JVM architecture? [3 M][CO1][BL1,BL2]

Q2 What will be the output of the following code snippet/program? Provide a step-by-step explanation of how the output is derived. [2 × 5 =10] [CO3,CO4][BL1, BL4]

a).
```
class Test {
    public static void main(String[] args) {
        int i = 5, j = 2;
        double d = i / j;
        System.out.print(d);
    }}
```

b).
```
int n = -8;
System.out.println(n >> 2);
```

c).
```
class Student {
    String name;
    Student(String n) { name = n; }
}
class Test {
    public static void main(String[] args) {
        Student[] s = new Student[2];
        System.out.println(s[0]);
    }
}
```

d).
```
class Test {
    static int x = 10;
    public static void main(String[] args) {
        Test t = null;
        System.out.println(t.x);
    }}
```

e).
```
class Test {
    void fun(long a) {
        System.out.println("long");
    }
    void fun(Integer a) {
        System.out.println("Integer");
    }}
public class Main {
    public static void main(String[] args) {
        Test t = new Test();
        t.fun(10);
    }}
```

Q3 a) Create a class Student with attributes name, roll number, and an array of marks. Implement a copy constructor that creates a deep copy of the marks array to ensure that changes to the copied object do not affect the original object. [4M][CO3, CO3][BL3,BL4]

b) Why can't a static method access instance variables or instance methods directly? How can a static method access instance members if needed? Illustrate your answer with a simple Java example. [2M][CO2][BL4]

c) Write a Java program that creates a HashMap to store the names and marks of five students. The program should display all students and their marks, update the marks of a student given their name, and display the total number of students who scored more than 75. [4M][CO3, CO4][BL3,BL4]